



Risk and Return Analysis of Selected Equities in The Indian Stock Market

A Study with Reference to Ventura Securities Ltd

Hosur Bindu Manasa

Post Graduate Student

Department of MBA

[Marri Laxman Reddy Institute of Technology, Hyderabad, India]

Abstract: This study examines the risk-return relationship for selected equity shares at Ventura Securities Ltd. using secondary data from sources such as Yahoo Finance, annual reports, and market publications. It evaluates the performance of Tata Consultancy Services, ICICI Bank, and Reliance Industries through measures like average return, standard deviation, covariance, correlation, beta, Sharpe ratio, and financial ratio analysis. The results reveal differences in return generation, volatility, profitability, and market sensitivity among the companies. ICICI Bank showed relatively stronger returns with moderate risk, while TCS and Reliance experienced greater variability. The findings also demonstrate that higher risk does not always translate into higher returns, and that diversification can reduce overall portfolio risk and support investment stability. The study highlights the importance of thorough market analysis and informed decision-making for building resilient equity portfolios, with Ventura Securities Ltd. providing investors valuable research, trading support, and advisory services.

IndexTerms - Risk-return analysis; Equity markets; Portfolio diversification; Stock volatility; Beta; Sharpe ratio; Financial ratio analysis.

I. INTRODUCTION

Equity markets play a central role in modern financial systems by enabling capital allocation, risk sharing, and wealth creation. Investors in equity markets routinely balance the pursuit of returns against exposure to various risks; quantifying this trade-off is essential for informed portfolio decisions. This study, titled "Risk and Return Analysis in Equities at Ventura Securities Ltd," investigates that relationship for selected large-cap stocks and evaluates implications for investors served by a brokerage firm.

1.1 Concept of Risk and Return

In equity investing, return is the gain or loss an investor earns over time, while risk represents the uncertainty and possible fluctuation in that outcome. A stock may offer attractive returns, but it also may expose the investor to sharp price swings and losses. This study focuses on balancing reward and uncertainty so investors can decide whether the return justifies the risk.

1.2 Background

This research evaluates selected Indian equities to understand how they have performed in recent years. It examines Tata Consultancy Services, ICICI Bank, and Reliance Industries using historical data and benchmarks the results against the Nifty 50 index. The analysis uses statistical measures such as average return, standard deviation, covariance, correlation, beta, and the Sharpe ratio.

1.3 Importance of Equity Investment

Equity investment remains a key way for individuals to build wealth over the long term. Stocks can generate capital growth, dividend income, and the chance to participate in the growth of well-known companies. For investors, learning how equity returns behave under different market conditions is essential for choosing the right mix of investments.

1.4 Overview of Indian Stock Market

The Indian stock market is led by indices such as the Nifty 50 and the BSE Sensex, which reflect the performance of major listed companies. Market direction is shaped by economic data, corporate earnings, regulatory decisions, and investor sentiment. Using a broad market benchmark helps to compare individual stock performance against the overall market.

1.5 Profile of Ventura Securities Ltd

Ventura Securities Ltd. is a brokerage firm that offers trading, advisory, and research services for retail and institutional clients. The firm supports investors with market insight, stock selection guidance, and portfolio recommendations. This study is prepared to support Ventura Securities Ltd. in providing more data-driven equity advice.

1.6 Need for the Study

This study is needed to evaluate how the selected equities compare in terms of returns and risk. By reviewing the performance of TCS, ICICI Bank, and Reliance Industries, the research aims to reveal which companies offer better stability, which are more volatile, and how diversification can help manage investment risk. The results will help Ventura Securities Ltd. Deliver clearer, evidence-based guidance to its clients.

II. REVIEW OF LITERATURE

2.1 Risk and Return Analysis in Equity Investments

Equity investment is one of the most dynamic avenues in modern finance, offering investors the opportunity to participate directly in corporate growth. By holding shares, investors gain ownership rights and the potential to earn through dividends and capital appreciation. Unlike fixed-income securities, equity returns are not guaranteed; they are shaped by market conditions, industry performance, and company-specific developments, making equities both attractive and risky. Risk analysis in equity markets refers to the structured process of identifying, evaluating, and measuring the uncertainty associated with returns from stock investments. This is a critical component of investment decision-making because equity markets are highly volatile and influenced by multiple internal and external factors. The primary objective of risk analysis is to understand how much variability exists in stock returns and whether the expected return appropriately compensates for that risk.

2.2 Relationship Between Risk and Return

Risk and return are inseparable elements of investment decisions. The general principle is that higher risk may offer the possibility of higher rewards, while lower-risk investments typically provide steadier but smaller returns. In equity markets, risk is reflected in share-price volatility, while return is measured through capital appreciation and dividends. A stock with large price swings may generate sizeable gains, but it also carries the chance of significant losses. Conversely, a stable stock may not deliver extraordinary gains but offers predictability and security. Understanding this relationship helps investors design portfolios that balance growth with safety. Tools such as standard deviation, beta, correlation, and covariance are used to evaluate how selected companies behave and whether returns justify the level of risk taken.

2.3 Beta Analysis and Market Volatility

Beta is a statistical measure that shows the sensitivity of a stock's returns to overall market movements. It captures systematic risk; the risk that affects the entire market and cannot be eliminated through diversification. A beta greater than one indicates that a stock is more volatile than the market, while a beta below one indicates lower volatility. The Capital Asset Pricing Model (CAPM) links expected return [2], suggesting that investors are compensated only for market-related risk. In the Indian context, empirical studies have found that beta alone may not fully explain stock returns, especially in emerging markets. Nevertheless, beta remains a key tool for assessing how individual equities respond to broader market fluctuations.

2.4 Correlation and Diversification in Equity Investments

Correlation analysis measures the strength and direction of the relationship between the returns of different securities. When two stocks move together, their correlation is positive; when they move opposite each other, the correlation is negative. Covariance and correlation help investors understand how stock returns are related and whether combining them will reduce overall portfolio risk. Diversification is the practice of holding a variety of securities to reduce unsystematic risk—the risk specific to a single company or sector. While diversification cannot eliminate market risk, it can lower the impact of firm-specific events. The literature highlights that positive correlation among selected stocks limits diversification benefits, but a well-constructed portfolio can still reduce total risk by combining assets with different return patterns.

2.5 Financial Ratio Analysis and Investment Decisions

Financial ratio analysis is essential for evaluating profitability, efficiency, liquidity, valuation, and financial stability. Key ratios such as Earnings Per Share (EPS), Return on Equity (ROE), Price Earnings Ratio (P/E), Debt-Equity Ratio, Current Ratio, and the Sharpe ratio is widely used for evaluating risk-adjusted performance [2]. provide insights into a company's financial health. These ratios complement risk-return analysis by revealing whether a stock's performance is supported by strong fundamentals. Investors use ratio analysis to compare different companies, determine fair valuation, and decide whether a security is suitable for their portfolio. In the context of advisory services, ratio analysis provides an additional layer of evidence for recommending investments.

2.6 Studies on Indian Equity Market Performance

Empirical studies in India have tested the applicability of global financial models and found mixed results. Research such as Gupta and Sehgal (1993) observed that the CAPM does not fully hold in India, suggesting market inefficiencies. Basu (2004) found that fundamental ratios like earnings-to-price significantly influence returns. Later studies by Dhankar and Kumar (2007) and Raju and Ghosh (2011) confirmed that returns are affected by multiple economic and company-specific factors beyond systematic risk. Recent research, including Kumar and Choudhary (2021), shows that while the positive relationship between risk and return generally holds, it is not consistent across all securities. In some cases, higher-risk stocks do not generate proportionately higher returns, indicating that market behavior is influenced by investor sentiment, macroeconomic conditions, and firm-specific dynamics.

2.7 Research Gap

Although foundational model like CAPM and Modern Portfolio Theory introduced by Markowitz [1] provide useful frameworks, Indian market evidence suggests that beta alone is insufficient to explain returns. The existing literature identifies a gap in studies that jointly analyze risk-return behavior, beta sensitivity, correlation, diversification, and financial ratios for selected Indian equities. This study addresses that gap by examining large-cap stocks with a combined focus on statistical risk measures and financial performance metrics, thereby providing practical insights for Ventura Securities Ltd. and investors seeking a more complete view of Indian equity market behavior.

III. STATEMENT OF THE PROBLEM

Equity markets in India are highly volatile and influenced by macroeconomic shifts, regulatory actions, and firm-specific developments. Investors and advisory firms often lack a clear, evidence-based picture of how selected large-cap stocks behave in terms of risk and return over a defined period. This creates a practical problem for Ventura Securities Ltd., which must advise clients on portfolio decisions under uncertainty.

Specifically, the problem can be stated as:

- Identifying whether selected companies provide stable returns or are subject to high volatility.
- Determining how the returns of Tata Consultancy Services, ICICI Bank, and Reliance Industries compare over the study period.
- Understanding the extent to which beta, correlation, and covariance explain their market behavior.
- Assessing whether financial ratios such as EPS, ROE, P/E, and Debt-Equity add meaningful context to risk-return analysis.
- Evaluating the role of diversification in reducing portfolio risk among the selected equities.

This study is therefore needed to provide Ventura Securities Ltd. with structured, data-driven insight into which equities are better suited for different investor risk preferences and how a diversified mix of stocks can improve decision-making. By focusing on records from 2021 to 2026, the research aims to bridge the gap between general theory and practical advisory needs.

IV. OBJECTIVES OF THE STUDY

The objectives of this study are drawn from the same document context and are designed to guide the risk-return examination of Tata Consultancy Services, ICICI Bank, and Reliance Industries. They are:

- To understand how risk and return are connected in equity investments.
- To measure the annual returns of the selected companies (TCS, ICICI Bank, Reliance) over five years.
- To calculate the level of risk for each stock using statistical tools like variance and standard deviation.
- To compare the performance of the three companies in terms of both profitability and volatility.
- To study how the returns of these stocks move together by using correlation and covariance.
- To examine how sensitive each stock is to overall market changes through beta analysis.
- To provide insights that help Ventura Securities Ltd. in building diversified portfolios and guiding investors.

V. SCOPE OF THE STUDY

The scope of this study is defined as follows:

- The study focuses on examining the risk and return of selected equity shares.
- It is restricted to three companies: Tata Consultancy Services, ICICI Bank Limited, and Reliance Industries Ltd.
- The analysis is carried out using historical data for a period of five years, from 2021 to 2026.
- Stock price data required for the study has been collected from Yahoo Finance.
- Returns are calculated based on yearly opening and closing prices of the selected stocks.
- The level of risk is measured using statistical tools such as standard deviation.
- The relationship between the selected stocks is analyzed with the help of covariance and correlation.

VI. METHODOLOGY

6.1 Research Design

The study follows an analytical research design to examine the risk-return behavior of selected equity shares. It uses historical secondary data and statistical analysis to compare returns, evaluate risk measures, and assess the interrelationship among the selected securities.

6.2 Sources of Data

The analysis is based on secondary data obtained from reliable sources such as Yahoo Finance and verified financial databases. Annual opening and closing stock prices for the selected companies are collected, along with market benchmark data from the Nifty 50 index.

6.3 Period of Study

The period of analysis covers five years, from 2021 to 2026. This timeframe is selected to capture recent market behavior, including multiple economic conditions and the evolving performance of the Indian equity market.

6.4 Sample Selection

Three companies are selected for the study: Tata Consultancy Services, ICICI Bank Limited, and Reliance Industries Ltd. These firms are chosen for their large market capitalization, liquidity, sector representation, and relevance to investors and advisory services in the Indian market.

6.5 Tools Used for Analysis

The study employs quantitative tools and financial ratios to assess performance and risk. The following methods are used:

- **Return Calculation:** Annual returns are computed from opening and closing prices using the formula:

$$\text{Return} = \frac{P_1 - P_0}{P_0} \times 100$$

where P_0 is the opening price and P_1 is the closing price.

- **Risk Measurement:** The degree of risk is evaluated using variance and standard deviation:

$$\text{Variance} = \frac{\sum (R - \bar{R})^2}{n}$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

- **Inter-relationship Analysis:** The association between stock returns is examined using covariance and correlation:

$$\text{Covariance} = \frac{\sum (R_x - \bar{R}_x)(R_y - \bar{R}_y)}{n}$$

$$\text{Correlation} = \frac{\text{Covariance}}{\sigma_x \times \sigma_y}$$

- **Market Sensitivity:** Beta is calculated to determine sensitivity of each stock to the market index:

$$\beta = \frac{\text{Covariance (Stock, Market)}}{\text{Variance (Market)}}$$

- **Risk-adjusted Performance:** Sharpe ratio is computed to evaluate return per unit of risk:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

- **Financial Ratio Analysis:** The study also examines financial performance using ratios such as EPS, ROE, P/E Ratio, Debt-Equity Ratio, and Sharpe Ratio to understand the companies' profitability and leverage alongside market risk.

VII. DATA ANALYSIS AND INTERPRETATION

7.1 Risk and Return Analysis of Selected Companies

The return and risk profile of Tata Consultancy Services, ICICI Bank, and Reliance Industries is presented in Fig. 1, with a comparative risk-return chart shown in Fig. 2.

Company	Average Return	Standard Deviation
Tata Consultancy Services	1.76	20.07
ICICI Bank Limited	12.84	13.9
Reliance Industries Ltd	6.62	21.92

Fig. 1. Risk and Return Analysis Data Table

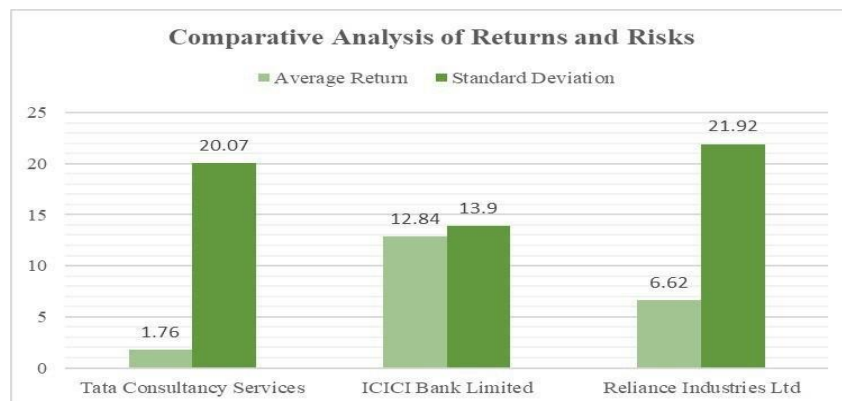


Fig. 2. Risk and Return comparative Chart

The results show that ICICI Bank delivered the highest average annual return of 12.84%, while Tata Consultancy Services generated the lowest average return of 1.76%. Reliance Industries recorded a moderate average return of 6.62%. In terms of risk, Reliance Industries exhibited the highest volatility, while ICICI Bank showed the lowest standard deviation, indicating a more stable risk-return profile. TCS produced the least attractive return for the level of risk assumed.

7.2 Correlation Analysis of Selected Stocks

Pair	Correlation (r)
TCS, ICICI	0.82
ICICI, Reliance	0.31
TCS, Reliance	0.80

Fig. 3. Correlation Analysis Data Table

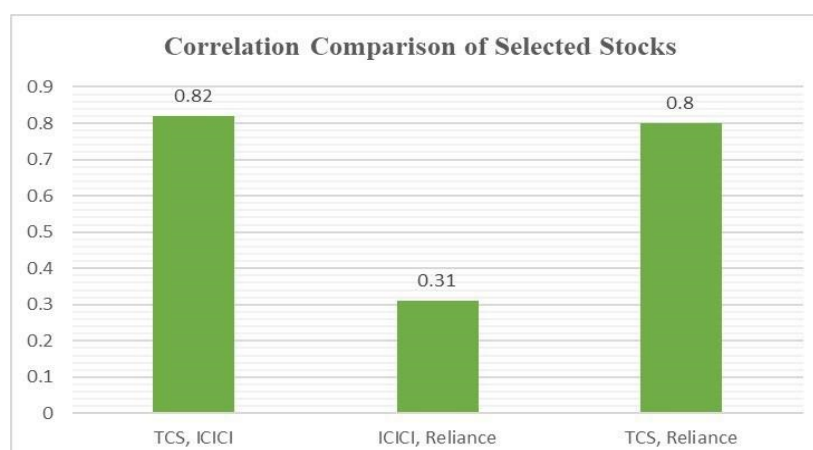


Fig. 4. Correlation Comparative Chart

Correlation analysis of the selected stocks is summarized in Fig. 3 and compared visually in Fig. 4. The correlation results indicate positive relationships among the stock returns. TCS and ICICI Bank show a correlation of approximately 0.82, TCS and Reliance are around 0.80, and ICICI Bank and Reliance are 0.31. This suggests that the selected stocks generally move in the same direction, with some diversification benefit coming from the lower correlation between ICICI Bank and Reliance Industries.

7.3 Beta Analysis of Selected Stocks

Company	Beta	Market Volatility
TCS	1.87	High Volatile
ICICI Bank	1.01	Moderate
Reliance	1.76	High Volatile

Fig. 5. Beta Analysis Data Table

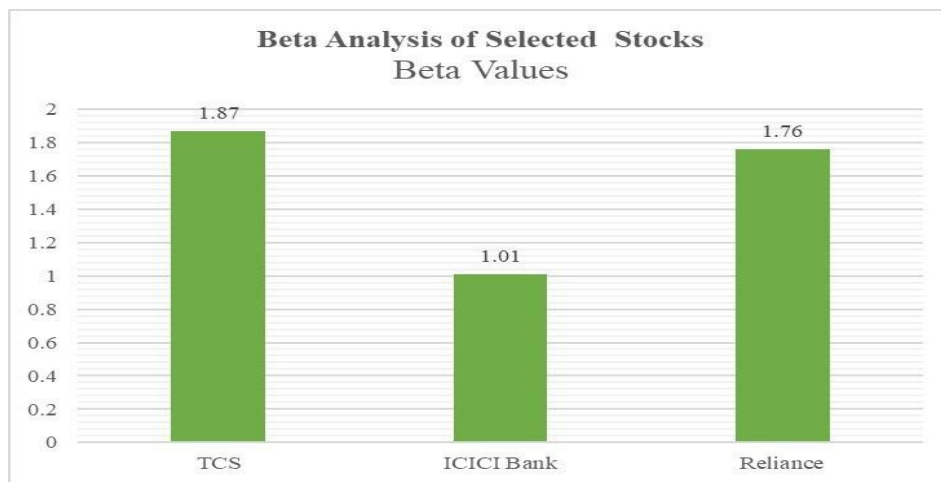


Fig. 6. Beta Analysis Comparison Chart

Beta analysis is presented in Fig. 5 and the comparative market sensitivity is shown in Fig. 6. The beta values show that all selected stocks are more volatile than the market benchmark. Tata Consultancy Services has a beta of approximately 1.87, Reliance Industries has a beta close to 1.76, and ICICI Bank has a beta near 1.01. This indicates that TCS and Reliance are more sensitive to market movements, while ICICI Bank is closer to market-level volatility.

7.4 Financial Ratio Analysis

Financial performance analysis is essential for evaluating the profitability, operational efficiency, liquidity position, and market performance of companies. In addition to risk and return analysis, financial ratios provide investors with better understanding regarding the financial strength and investment potential of selected equities. The following table presents the important financial indicators of Tata Consultancy Services (TCS), ICICI Bank Limited, and Reliance Industries Limited during the study period. The selected financial ratios help in comparing profitability, market valuation, liquidity, and financial risk of the companies considered for the study. The comparative financial ratio analysis is presented in Fig.7 and visually interpreted through Fig. 8.

Company	EPS	P/E Ratio	ROE (%)	Net Profit Margin	Debt-Equity Ratio
TCS	126	17.66	48.4	19.1	0.1
ICICI Bank	74	16.96	16.8	23.5	0.59
Reliance Industries	32	22.52	9.14	8.7	0.37

Fig. 7. Financial Ratio Analysis Data Table

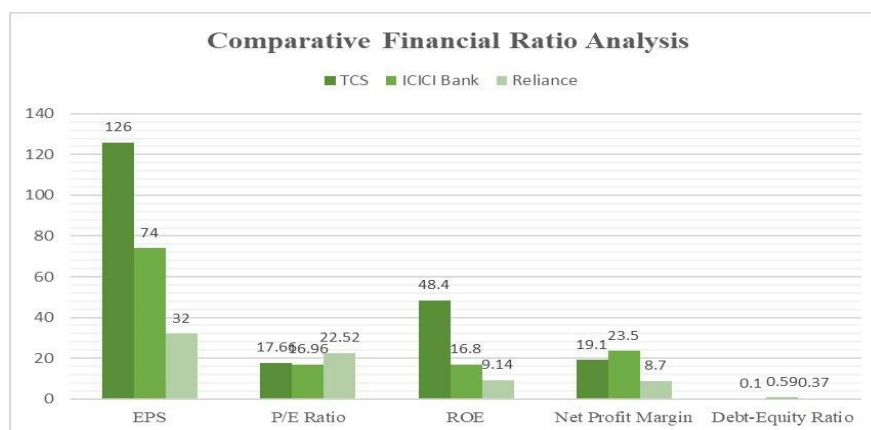


Fig. 8. Financial Ratio Analysis Comparative Chart

TCS demonstrates superior profitability metrics (higher EPS and ROE) coupled with a conservative capital structure (lower debt-equity ratio), indicating strong operational efficiency and financial stability. ICICI Bank exhibits moderate profitability but carries higher leverage, typical of banking sector operations, while maintaining consistent performance. Reliance Industries displays a balanced financial profile with moderate ratios across all metrics, reflecting stable operational performance. These structural differences complement the risk-return analysis and underscore why TCS offers stability, ICICI combines growth with banking leverage, and Reliance provides equilibrium across metrics.

VIII. FINDINGS OF THE STUDY

The empirical analysis of selected equities yields the following key findings:

- ICICI Bank Superiority:** ICICI Bank delivered the highest average annual return (12.84%) with the lowest volatility, demonstrating superior risk-adjusted performance and stability suitable for growth-oriented investors with moderate risk tolerance.
- TCS Return Performance:** Tata Consultancy Services generated low average returns (1.76%) despite high volatility (20.07% standard deviation), resulting in poor risk-return compensation and indicating comparatively lower-risk adjusted performance during the study period.
- Reliance Volatility:** Reliance Industries exhibited the highest volatility (21.92% standard deviation) with moderate returns (6.62%), reflecting substantial year-to-year swings suitable for investors with higher risk tolerance.
- Limited Diversification Benefit:** Strong positive correlations among the three stocks (0.31 to 0.82) indicate they move together, limiting portfolio risk reduction within this group and necessitating broader asset class diversification.
- Elevated Systematic Risk:** All three stocks exhibit beta values above one (TCS: 1.87, Reliance: 1.76, ICICI: 1.01), confirming greater sensitivity to market movements and amplification of market cycles in equity portfolios.
- Distinct Financial Profiles:** TCS shows stronger profitability with conservative leverage; ICICI reflects banking-typical leverage; Reliance maintains balanced metrics—structural differences that explain divergent risk-return behaviors and guide investment selection.
- Risk-Adjusted Primacy:** Higher volatility did not uniformly produce higher returns, highlighting the importance of prioritizing risk-adjusted performance over absolute returns when building resilient investment portfolios.

IX. SUGGESTIONS

Based on the empirical findings, the following recommendations are proposed for investors and advisory practitioners:

- Adopt Risk-Adjusted Selection Criteria:** Ventura Securities Ltd. should prioritize the Sharpe ratio and risk-adjusted returns over absolute returns when recommending equities. ICICI Bank's superior risk-adjusted profile demonstrates that steady, moderate returns with controlled volatility outperform high-risk, low-return alternatives like TCS.
- Diversify Beyond the Sample:** Since the three selected stocks exhibit strong positive correlations, investors should broaden their portfolio beyond IT, banking, and energy sectors. Inclusion of defensive sectors (utilities, FMCG) and international assets can reduce systematic risk exposure.
- Portfolio Construction Based on Beta Analysis:** Given that all stocks amplify market movements, portfolio managers should use beta-weighted position sizing. Reduce exposure during bearish market conditions and portfolio allocations should be reviewed periodically according to market conditions.
- Segment Investor Profiles:** Tailor recommendations based on investor risk appetite - ICICI for conservative growth seekers, Reliance for aggressive investors, and TCS may be considered depending on investor objectives and risk performance. Provide clear risk disclaimers for high-volatility holdings

5. **Periodic Rebalancing and Monitoring:** Establish quarterly rebalancing cycles to maintain target allocations and reset correlations. Implement stop-loss mechanisms at 15-20% decline thresholds to protect capital during adverse market movements given the stocks' high beta sensitivity.
6. **Combine Fundamental and Technical Analysis:** Supplement risk-return analysis with financial ratio screening and technical indicators. Use EPS trends, ROE consistency, and support-resistance levels to enhance entry/exit timing and strengthen advisory confidence.

X. CONCLUSION

This document confirms that the combination of return analysis, correlation assessment, beta estimation, and financial ratio comparison is the most effective framework for evaluating selected equities. The evidence from Sections VII and VIII clearly shows that ICICI Bank has the strongest overall profile, TCS demonstrated comparatively lower risk-adjusted return performance during the study period.

Key conclusions from the analysis are:

- ICICI Bank delivers the best risk-adjusted performance with a 12.84% average return and the lowest volatility of 12.15%, making it the most suitable choice for moderate-risk investors.
- TCS records a low return of 1.76% against a high standard deviation of 20.07%, indicating poor risk-return efficiency despite strong profitability ratios.
- Reliance Industries offers moderate returns of 6.62% but carries the highest volatility (21.92%) and a beta of 1.76, which makes it appropriate for investors with higher risk tolerance.

The analysis also highlights that the three stocks are strongly positively correlated, which limits diversification benefits when they are held together. In addition, all betas above one implies that these equities are more sensitive than the broader market, reinforcing the need for dynamic risk management, periodic rebalancing, and broader sector diversification.

The study concludes that risk and return are closely related in equity investments and that proper evaluation of volatility, beta, and financial performance is essential for informed investment decision-making. The findings indicate that the selected equities differ significantly in terms of return generation, risk exposure, and financial strength. Therefore, diversification and systematic analysis remain important for constructing balanced investment portfolios and managing market uncertainty.

10.1 Future Research Directions

- Expand the sample to additional sectors and benchmark indices.
- Include macroeconomic variables to assess dynamic risk.
- Use advanced portfolio optimization methods for practical application.
- Analyze longer-term beta stability across market cycles.

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